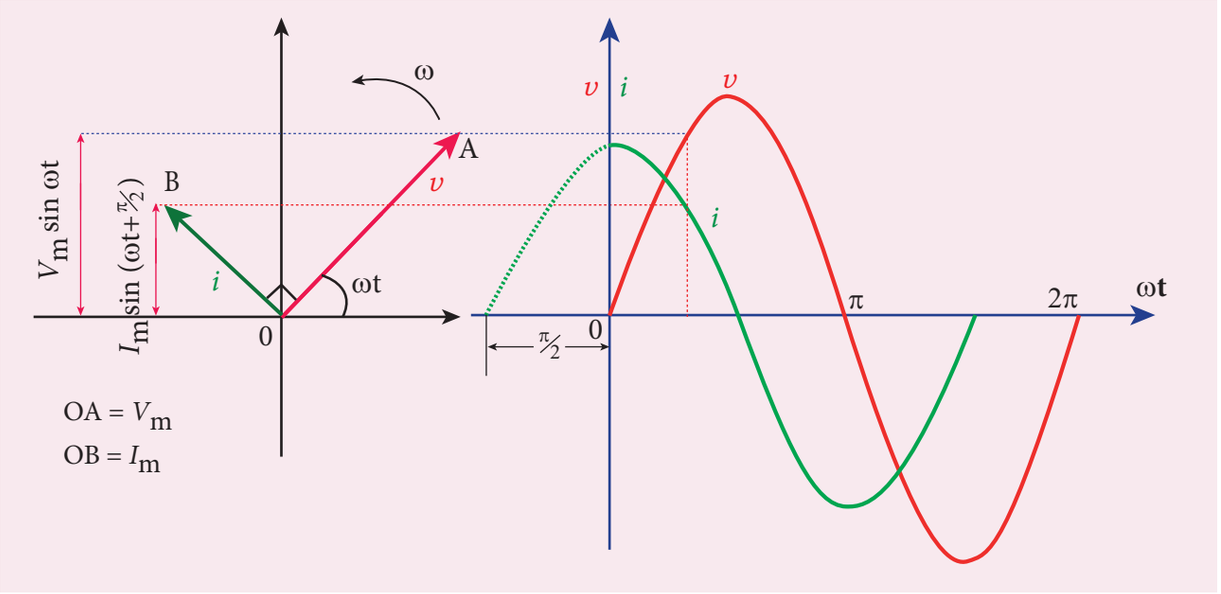
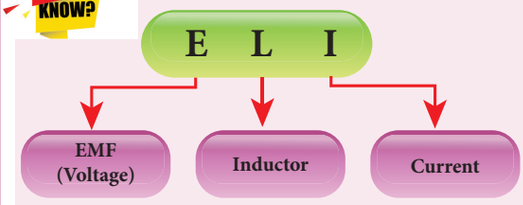


Figure 4.45 Phasor diagram and wave diagram for AC circuit with C

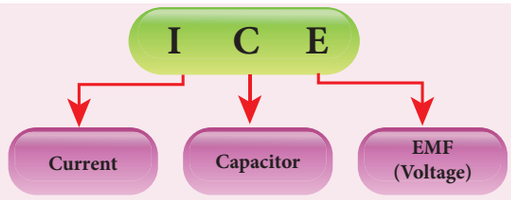


What is ELI?



ELI is an acronym which means that EMF (voltage) leads the current in an inductive circuit.

What is ICE?



ICE is an acronym which means that the current leads the EMF (voltage) in a capacitive circuit.

EXAMPLE 4.20

A 400 mH coil of negligible resistance is connected to an AC circuit in which an effective current of 6 mA is flowing. Find out the voltage across the coil if the frequency is 1000 Hz.

Solution

$L = 400 \times 10^{-3}$ H; $I_{eff} = 6 \times 10^{-3}$A

$f = 1000$ Hz

Inductive reactance, $X_L = L\omega = L \times 2\pi f$

$= 2 \times 3.14 \times 1000 \times 0.4$

$= 2512\ \Omega$

Voltage across L,

$V = I X_L = 6 \times 10^{-3} \times 2512$

$V = 15.072\,\text{V}(RMS)$

EXAMPLE 4.21

A capacitor of capacitance $\frac{10^2}{\pi}\mu$F is connected across a 220 V, 50 Hz A.C. mains. Calculate the capacitive reactance, RMS value of current and write down the equations of voltage and current.

Solution

$C = \frac{10^2}{\pi} \times 10^{-6}$ F, $V_{RMS} = 220$ V; $f = 50$ Hz

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